

MOUNT EDGECOMBE, South AfricaWMO: **685830**Lat: **29.7067S**Lon: **31.0461E**Elev: **308**StdP: **14.53**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	49.2	51.2	34.4	29.4	60.0	37.7	33.6	59.1	15.1	66.3	13.6	67.0	1.5	260	0.261

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.4	86.7	74.5	84.7	73.8	83.0	73.1	76.6	83.6	75.6	82.2	74.7	81.0	6.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	130.6	80.9	73.5	126.0	80.2	72.5	121.8	79.3	40.4	83.8	39.3	82.4	38.4	81.3	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.2	11.5	46.2	93.1	3.1	3.0	43.9	95.2	42.1	96.9	40.4	98.6	38.2	100.8		
(5)			42.1	79.1	3.1	2.8	39.9	81.1	38.1	82.7	36.4	84.2	34.2	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	75.0	75.7	74.1	70.6	67.2	64.0	62.8	64.9	66.3	68.1	70.1	72.8	(6)
(7)		DBStd	5.70	3.56	3.37	3.89	3.76	3.89	3.37	3.38	3.73	4.35	4.07	4.07	4.08	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	272	0	0	1	4	21	56	84	47	34	17	6	2	(9)
(10)		CDD50	7031	774	719	748	618	534	419	396	461	489	561	604	708	(10)
(11)		CDD65	1829	309	299	284	173	89	25	15	43	74	113	160	245	(11)
(12)		CDH74	8532	1836	1797	1489	550	287	97	82	135	224	310	577	1148	(12)
(13)		CDH80	1380	320	329	230	68	43	14	12	17	37	47	80	183	(13)
(14)	Wind	WSAvg	4.4	5.0	4.5	4.2	3.9	3.4	3.3	3.5	4.3	4.9	5.4	5.3	5.1	(14)
(15)	Precipitation	PrecAvg	33.8	4.7	3.5	3.4	2.0	1.3	0.8	1.7	1.3	2.3	3.9	3.9	4.9	(15)
(16)		PrecMax	45.4	11.9	9.0	8.1	5.4	5.7	2.7	7.6	4.3	7.9	10.7	10.4	12.6	(16)
(17)		PrecMin	16.6	1.9	0.6	1.1	0.4	0.0	0.0	0.0	0.1	0.5	0.8	1.3	1.6	(17)
(18)		PrecStd	7.8	2.2	2.3	2.0	1.3	1.4	0.8	2.4	1.0	1.5	2.1	2.0	2.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	88.5	87.3	85.7	84.8	81.8	81.6	81.9	84.5	85.7	86.5	88.2	(19)
(20)			MCWB	76.1	77.2	75.9	71.9	67.4	62.8	61.1	64.8	67.8	72.1	74.6	76.7	(20)
(21)		2%	DB	85.3	85.7	84.5	81.6	79.8	76.4	75.9	77.3	79.5	80.4	82.0	83.9	(21)
(22)			MCWB	75.0	75.4	74.5	71.3	67.6	61.8	61.5	63.8	67.7	69.3	72.0	74.2	(22)
(23)		5%	DB	83.3	83.6	82.5	79.2	76.9	73.4	72.6	74.5	76.2	77.1	79.4	81.7	(23)
(24)			MCWB	74.2	74.5	73.7	70.3	66.4	62.0	61.0	63.5	66.3	68.1	71.0	73.0	(24)
(25)		10%	DB	81.5	81.8	80.7	77.1	74.4	70.9	70.0	72.1	73.6	74.8	77.2	79.6	(25)
(26)			MCWB	73.3	73.5	72.8	69.3	65.7	61.1	60.2	62.6	65.2	67.2	69.8	72.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	78.4	77.2	74.2	72.0	67.1	65.8	68.3	70.9	72.7	75.5	77.3	(27)
(28)			MCD B	85.0	86.2	84.9	82.3	80.0	74.4	74.3	77.3	79.4	82.6	83.7	85.5	(28)
(29)		2%	WB	76.1	76.6	75.7	72.6	69.7	65.1	64.1	66.5	68.8	70.7	73.3	75.4	(29)
(30)			MCD B	83.1	83.0	82.4	79.4	76.9	72.4	71.8	73.7	76.4	77.7	79.8	81.7	(30)
(31)		5%	WB	75.0	75.5	74.6	71.3	68.0	63.7	62.7	65.2	67.4	69.4	71.9	74.2	(31)
(32)			MCD B	81.6	81.8	80.8	77.3	74.5	70.8	69.9	72.0	74.2	75.3	77.7	80.0	(32)
(33)		10%	WB	74.0	74.5	73.5	70.1	66.6	62.2	61.3	63.9	66.1	68.2	70.8	72.9	(33)
(34)			MCD B	80.3	80.5	79.6	75.7	72.7	69.1	68.0	70.5	72.4	73.8	76.2	78.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.4	11.4	11.6	12.8	15.6	17.6	17.5	15.3	12.8	11.3	11.1	11.0	(35)
(36)			MCDBR	14.9	14.3	14.1	16.0	19.0	21.1	21.8	19.7	18.5	16.5	15.6	14.7	(36)
(37)			MCWBR	7.4	7.1	7.0	7.9	9.8	11.6	11.9	10.0	9.7	9.2	8.8	7.8	(37)
(38)		5% WB	MCDBR	13.9	13.2	13.3	14.3	16.3	18.1	19.1	16.2	16.1	14.2	13.7	13.5	(38)
(39)			MCWBR	7.3	6.8	7.1	7.6	9.3	11.2	11.7	9.1	9.0	8.4	8.0	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.453	0.442	0.435	0.404	0.378	0.362	0.372	0.435	0.485	0.497	0.468	0.461	(40)	
(41)		taud	2.303	2.338	2.340	2.386	2.424	2.445	2.404	2.216	2.099	2.110	2.225	2.264	(41)	
(42)		Ebn at Noon	283	280	272	263	255	252	254	249	252	262	277	282	(42)	
(43)		Edh at Noon	44	42	39	34	30	27	30	40	49	52	48	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1879	1844	1606	1305	1118	995	1062	1238	1426	1556	1715	1853	(44)	
(45)		RadStd	149	137	80	81	56	47	70	75	91	139	144	136	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page